

面向设计前期

FOR EARLY-STAGE DESIGN

# 轻量 BIM Agent

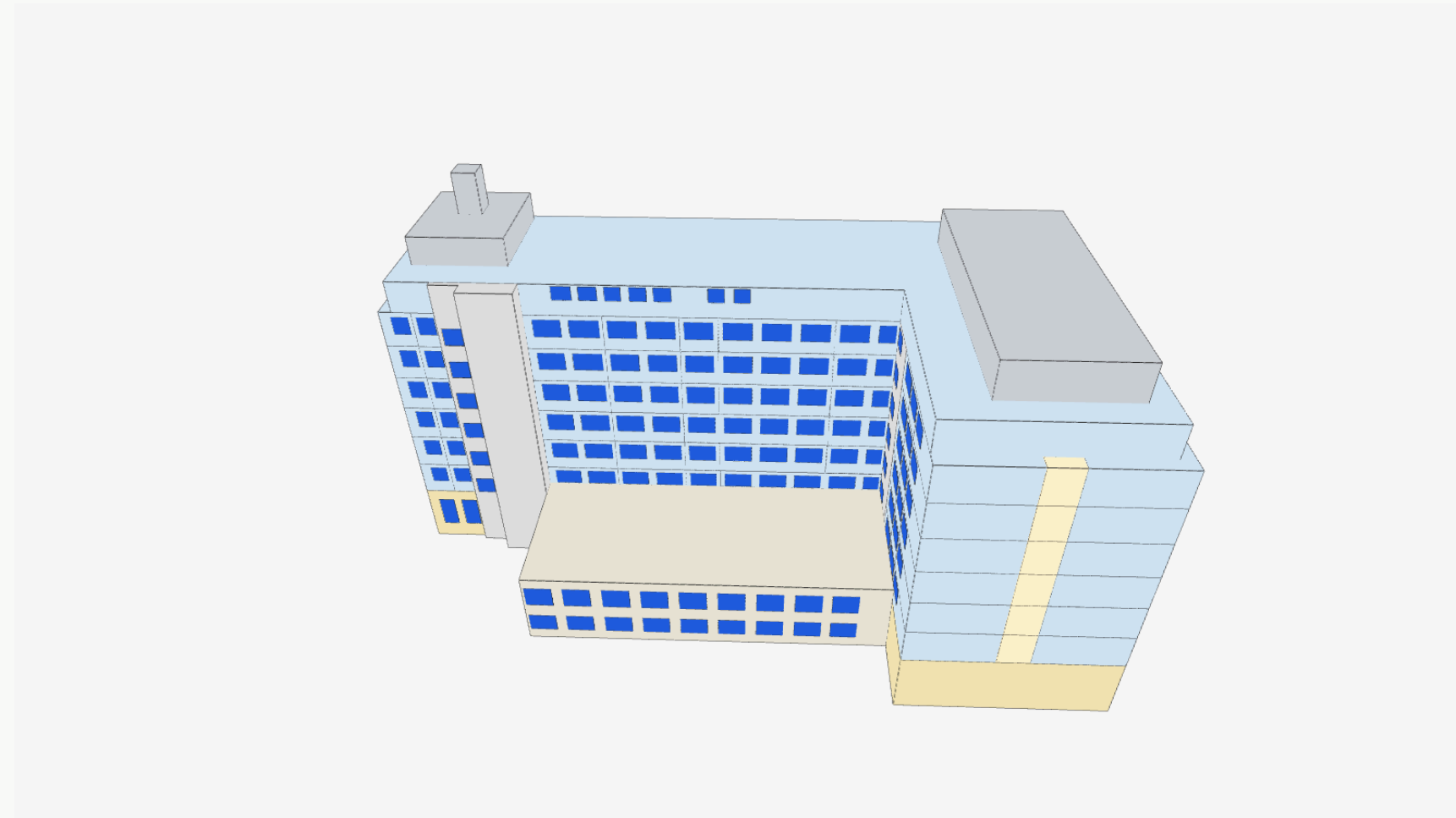
A lightweight BIM agent  
for building performance simulation

让建筑仿真更早参与方案设计

Bringing simulation into early design decisions

多模态输入 → 轻量 BIM → 多物理场仿真

Multimodal inputs · Lightweight BIM · Multiphysics simulation

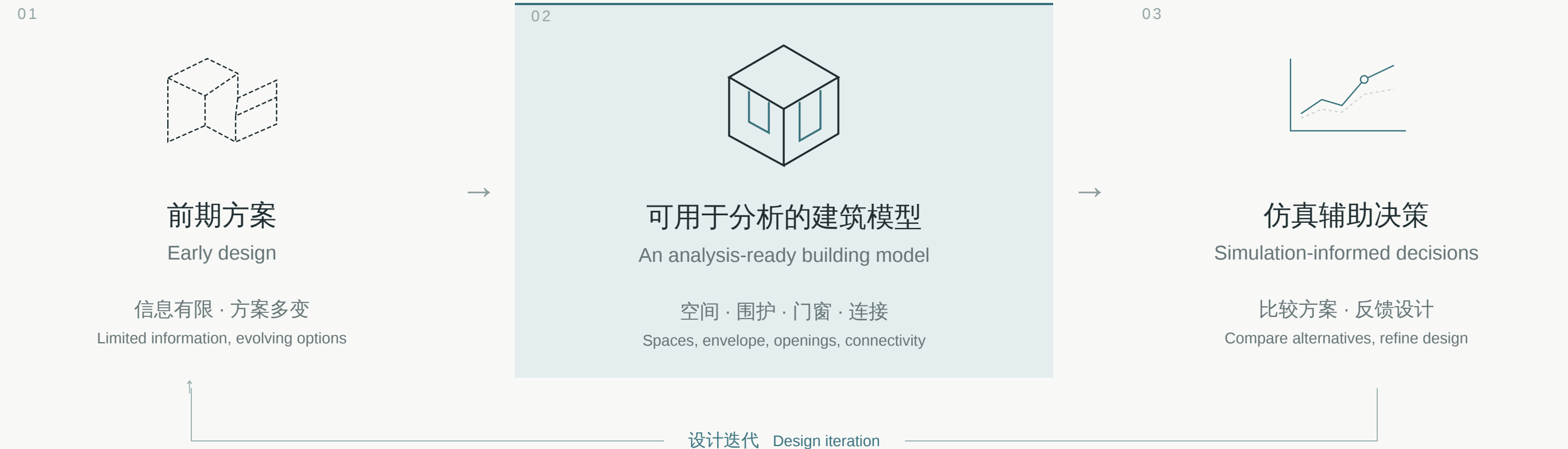


■ VOIMATALO

轻量 BIM 探索 / BIM exploration

# 在设计前期，如何支撑相对高精度的建筑仿真？

How can early-stage design support higher-fidelity building simulation?



## 研究切入点：提高前期建筑模型的信息质量

Focus: improve the information quality of early-stage building models

相对高精度为研究目标，尚待系统对照验证。 Higher fidelity is a research objective; systematic evaluation is pending.

# 多模态输入，共同的轻量 BIM 表达

Multimodal inputs, one shared lightweight BIM representation

前期输入 Early inputs

图纸 / Drawings

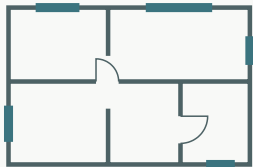
图像 / Images

贴图体量 / Textured meshes

意图 / Design intent

已知信息较多 More known information

推理需求增加 Greater need for inference →

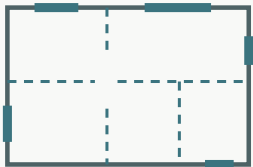


I

## 还原建模

Reconstruction

依据较充分的信息还原空间  
Recover spaces from sufficient evidence

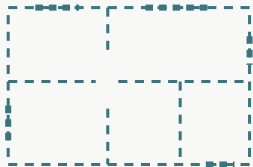


II

## 部分推理建模

Partial inference

保留已知约束，推断缺失部分  
Retain evidence, infer what is missing



III

## 完全推理建模

Full inference

从需求与设计意图生成空间  
Generate spaces from requirements and intent

## 分类依据是信息充分程度

Categories reflect information sufficiency

—— 已知 / Known    - - - 推断 / Inferred

# Agent 组织建模，轻量 BIM 连接仿真

Agent-led modelling, a shared BIM for simulation backends



## 保留建筑空间，按后端需求派生计算模型

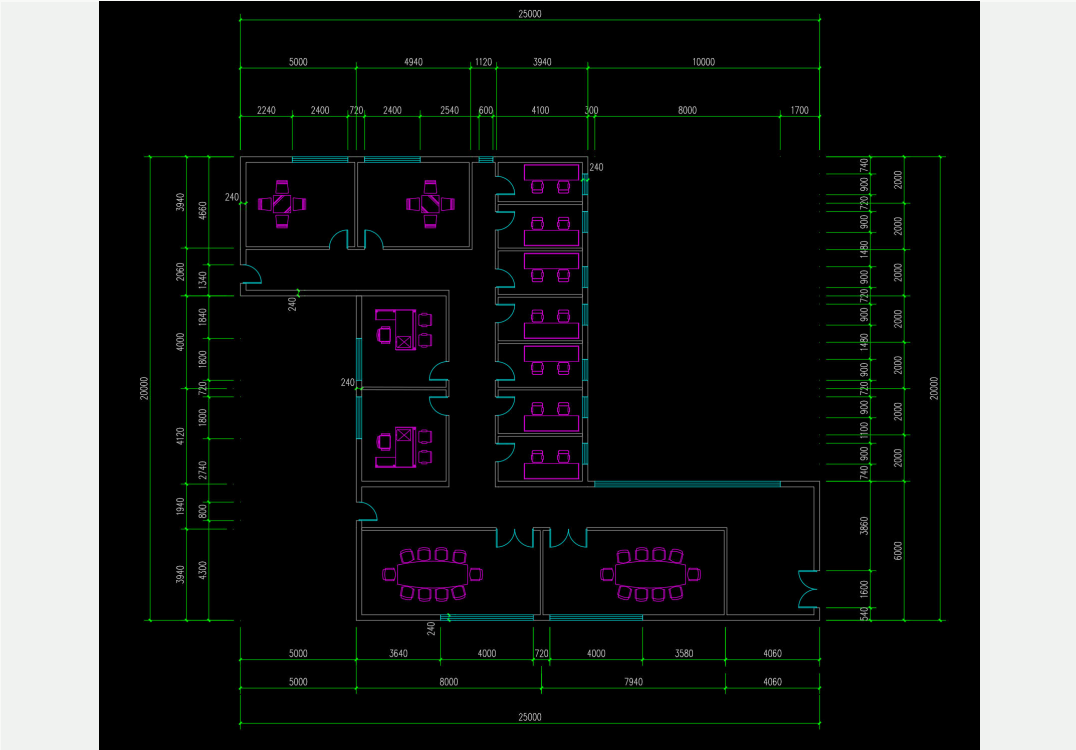
Preserve building spaces; derive simulation-specific models

EnergyPlus 为已有下游通路；本次两份展示未新增仿真验证。    EnergyPlus is an existing integration; these two demos have no new simulation validation.

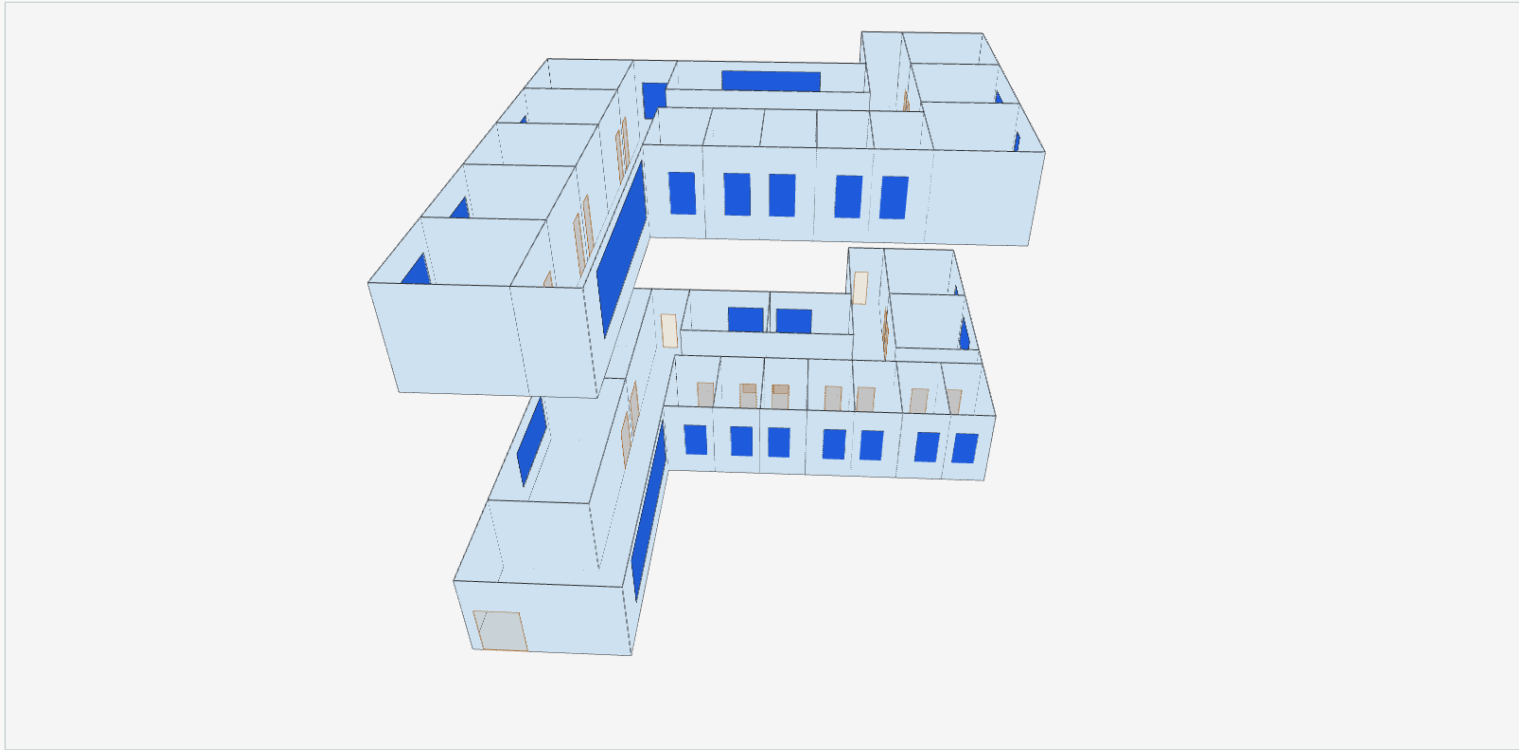
# 从建筑图纸到可查看的空间模型

From architectural drawings to an inspectable spatial model

输入 / INPUT sm25 · 一层平面  
Ground-floor plan



输出 / OUTPUT sm25 · 轻量 BIM  
Lightweight BIM



29 空间 Spaces  
31 窗 Windows  
30 门 Doors

旋转 · 分层 · 展开  
Rotate · Isolate floors · Explode

展示副本含人工修订，用于效果示意。 Assisted showcase with manual corrections.

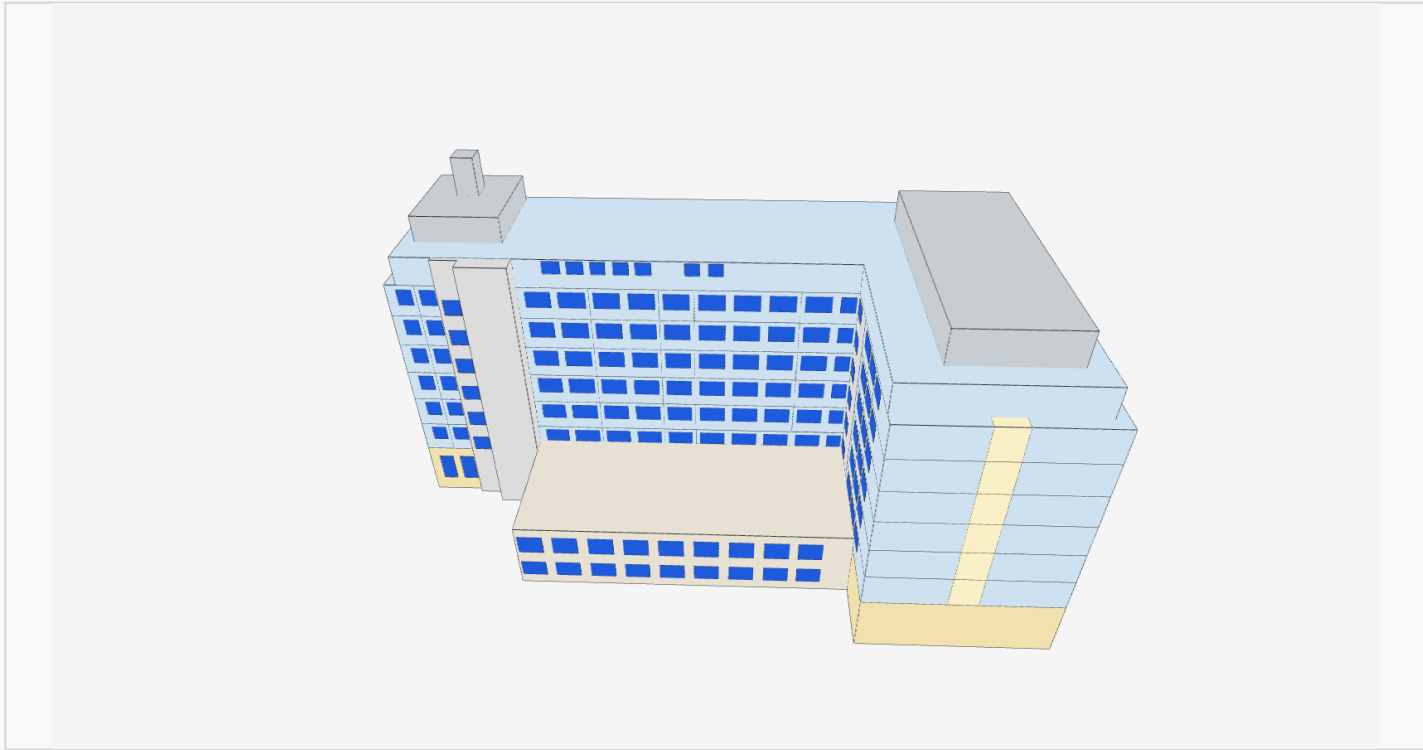
# 从真实贴图体量到轻量 BIM

From a real textured mesh to lightweight BIM

输入 / INPUT    Voimatalo · Helsinki  
Photogrammetry · GLB



输出 / OUTPUT    空间与门窗表达  
Spaces & openings



8 主楼层数    322 窗组  
Main storeys    Window groups

补全缺失体量，推断内部空间  
Complete missing geometry; infer interior spaces

主助手辅助探索；内部布局为推断，未核实。来源：SUM / Helsinki.    Assistant-led exploration; interiors are inferred and unverified. Source: SUM / Helsinki.

# 从可查看的模型，走向可靠的设计工具

From inspectable models to a reliable design tool

## 01 提升空间理解能力

Stronger spatial understanding

空间、连接与建筑逻辑  
Spaces, connectivity & architectural logic

## 02 提高复杂建筑稳定性

Robustness on complex buildings

跨案例、跨形态的可靠表达  
Reliable modelling across cases & forms

## 03 降低模型依赖度

Less reliance on large models

工具支撑与轻量模型协同  
Geometry tools & smaller-model collaboration

## 04 轻量 BIM 交互式编辑

Interactive BIM editing

查看、修改、反馈设计  
Inspect, edit & feed back into design

## ■ 服务于设计前期的多物理场仿真 Enabling multiphysics simulation in early-stage design